

Zhongyuan Zhao

Curriculum Vitae

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Keywords

Networked AI Systems	Machine Learning on Graphs
Self-Organizing Networks	Neural-Algorithmic Methods
Stochastic Network Optimization	Combinatorial Optimization
Signal Processing for Wireless Systems	Model-Informed Deep Learning

Education

- 2019 **Ph.D. in Computer Engineering**, *University of Nebraska-Lincoln*.
Minor in Finance (15 credits)
Advisor: Mehmet C. Vuran
- 2009 **M.S. in Signal and Information Processing**, *University of Electronic Science and Technology of China*, Department of Electronic Engineering.
Advisor: Zishu He
- 2006 **B.S. in Information Countermeasure Technology**, *University of Electronic Science and Technology of China*, Department of Electronic Engineering.

Research Experience

- 2024–Cont. **Assistant Research Professor**, RICE UNIVERSITY, Houston, TX.
- 2019–2024 **Postdoctoral Research Associate**, RICE UNIVERSITY, Houston, TX.
Conduct research on graph neural networks and graph-based reinforcement learning to address the resource allocation challenges in infrastructureless wireless networks, in particular, seeking distributed solutions for near-optimal link scheduling, routing, and computational offloading.
Write research papers, reports, and proposals. Supervise PhD and undergraduate students.
- 2019 **Software Developer**, UNIVERSITY OF NEBRASKA-LINCOLN, Lincoln, NE.
Developed software with Python+R to facilitate drug repurposing for gene-related diseases.
- 2013–2019 **Research Assistant**, UNIVERSITY OF NEBRASKA-LINCOLN, Lincoln, NE.
Conducted research on smart wireless networks for enhanced network capacity and spectrum efficiency. Developed interactive spectrum sharing, large-scale wireless testbed, and artificial neural network-based wireless receiver. Wrote research papers, reports, and proposals.
- 2011–2013 **Radio-Frequency Integration and Verification Engineer**, ERICSSON, Chengdu, China.
Conducted R&D tests for radio frontends in TD-LTE base-stations. Led radio test automation.
- 2009–2011 **DSP Software Engineer**, ARRAYCOMM, Chengdu, China.
Developed digital signal processing (DSP) software for 4th generation (4G) cellular base-stations.

2006–2009 **Research Assistant**, UNIVERSITY OF ELECTRONIC SCIENCE AND TECHNOLOGY OF CHINA, Department of Electronic Engineering, Chengdu, China.
Designed mixed-signal printed circuit boards (PCB) and developed DSP and FPGA software for a software-defined radio receiver and a PID control system for high-power electrical motor.

Publication Summary

Journal 10 peer-reviewed journal articles since 2013 (6 as first author) in wireless & network systems (TWC, JSAC, TMLCN, TVT, CC, Ad-Hoc), operations research (M&SOM), bioinformatics
Conference 19 peer-reviewed conference papers (13 as first author) in machine learning (ICLR), signal processing (ICASSP, CAMSAP), wireless systems (MILCOM, GlobeCom, DySPAN, BalkanCom)
Patent 2 patents (1 as first author) in digital signal processing and analog circuit design
Impact 686 citations, 12 h-index, 15 i10-index (Google Scholar, 08/29/2026)

[Full list of publications can be found at the end of this CV](#)

Research Proposal Developments

2025 Lead PI, “Scalable Context-aware Networking for Extremely Heterogeneous Networks” to Army Research Office, amount: \$661,102, Co-PI: Santiago Segarra
2023 Participated in writing proposal “Distributed Machine Learning for Tactical Networks” to Army Research Office, award amount: \$1.1M
2015–2018 Participated in writing proposal “CoSeC-RAN: Cognitive Secure Cloud RAN for Efficient Spectrum Sharing”, award: NSF CNS #1731833, amount \$435,399; and an US-Ignite grant.

Teaching Experience

2023–Cont. **PhD Research Mentor**, *Dept. of Electrical and Computer Engineering*, Rice University.
Advise three PhD students in research [C1, M1, R1, R2, C2, C3, C5, C7].
2020–Cont. **Undergraduate Research Mentor**, *Dept. of Electrical and Computer Engineering*, Rice University.
Guided 5 undergraduates in 3-month research internships (Summer 2020, 2022 [C10], 2025, 2026).
Served as research mentor in Summer Undergraduate Research Fellowships (SURF) 2025.
2017–2019 **Teaching Assistant**, *Dept. of Computer Science & Engineering*, Univ. of Nebraska-Lincoln.
Data Structure and Algorithms (Fall 2017, Spring 2019). Multi-Agent System (Fall 2017).
2017 **ITA Teaching Certificate**, *Institute for International Teaching Assistants*, University of Nebraska-Lincoln.
Advanced training for teaching on the college and university level, August 2017.
2016–2018 **Senior Design Mentor**, *Jeffrey S. Raikes School of Computer Science and Management*, University of Nebraska-Lincoln.
Guided undergraduate teams in their senior design projects (2016-2017 [C17], 2017-2018).
2005–2007 **Teaching Assistant**, *Undergraduate Innovation Centre*, University of Electronic Science and Technology of China.
Advised undergraduates in a 2-year training program for national electronic design contest.

Selected Projects

- 2026–Cont. **Scalable Context-aware Networking for Extremely Heterogeneous Networks**, *Rice University*, PI: Zhongyuan Zhao, Co-PI: Santiago Segarra, refs: [C1].
Investigate scalable intelligent networking methods for tactical networks with heterogeneous wireless technologies by combining stochastic network optimization and graph-based machine learning. Develop distributed solutions for joint routing, scheduling, and modality selection under interference and dynamic network conditions, optimizing multiple objectives including throughput, latency, resiliency, security, and resource efficiency.
- 2023–2026 **Distributed Machine Learning for Tactical Networks**, *Rice University*, PI: Santiago Segarra, Co-PI: Ashutosh Sabharwal, refs: [M1, C1, R2, R1, C2, C3, C5, C6, C7, C8, J2].
Investigate distributed multi-hop computational offloading in infrastructureless wireless networks to support the army's multi-domain operations through edge AI. Develop graph neural networks, distributed learning, and fail-safe mechanism to enhance the contextual awareness, adaptability, scalability, and robustness of edge AI solutions in self-organizing autonomous networks.
- 2019–2024. **Autonomous Networking for Multi-domain Operations**, *Rice University*, PI: Ashutosh Sabharwal, Edward W. Knightly, Santiago Segarra, refs: [C9, C10, C11, C12, C13, C14, J2, J3, J5, J1].
Conduct research on graph neural networks and graph-based reinforcement learning to address resource allocation challenges in infrastructureless wireless networks, in support of the army's multi-domain operations. Develop efficient and distributed heuristics for link scheduling, routing, and computational offloading, which can improve the efficiency, performance, and edge intelligence of networks while retaining their self-organizing capabilities, scalability, and robustness.
- 2019 **Computational Biology for Drug Repurposing**, *University of Nebraska-Lincoln*, PI: Tomáš Helikar, references: [J4].
Developed software with Python+R to facilitate identification of potential drug candidates for gene-related diseases, by analyzing human genome data and molecular interaction networks.
- 2018–2021 **Neural OFDM Receiver**, *University of Nebraska-Lincoln*, PI: Mehmet C. Vuran, co-PI: Stephen D. Scott, references: [J6, T1].
Developed a wireless receiver using deep neural networks (DNNs) to directly process time-domain orthogonal frequency-division multiplexing (OFDM) signals, gaining extra performance from redundancy in the cyclic prefix (CP) discarded by FFT/iFFT-based receivers. It established a data-driven model-inspired approach for addressing signal artifacts and model mismatches in wireless systems.
- 2017 **Experimental Vehicle-to-Barrier Communications**, *University of Nebraska-Lincoln*, PI: Mehmet C. Vuran, references: [J10].
Investigated transient behaviors of wireless channels in vehicle-to-barrier communications under different antenna configurations, in order to provide design guidelines for smart barriers for reducing single-vehicle crashes. Participated in measurements and led the data analysis of 5.8GHz radio propagation between vehicles and road-side barriers during experimental crash tests.
- 2017–2019 **Nebraska Experimental Testbed of Things (NEXTT)**, *University of Nebraska-Lincoln*, PI: Mehmet C. Vuran, references: [C15, J7, T1].
Collaborated with university, government, and industrial partners to develop NEXTT, a large-scale cloud-radio access network (Cloud-RAN) testbed. NEXTT connects computing clusters to software-defined radio (SDR) transceivers deployed across two university campuses and a public street. It facilitates wireless research and education by providing researchers and students with easy access to live experiments and datasets in areas such as cellular, vehicle-to-infrastructure, and underground-to-aboveground communications. The testbed has benefited hundreds of researchers and students, and supported multiple courses at Univ. of Nebraska-Lincoln (CSCE 465, 422, 990, ECE6102).

- 2013–2017 **Cog-TV: Business and Technical Analysis of Cognitive Radio TV Sets for Enhanced Spectrum Access**, *University of Nebraska-Lincoln*, PI: Mehmet C. Vuran, co-PI: Demet Batur, references: [C16, C17, C18, J9, J8, T1].
Studied the feasibility of providing wireless broadband services on the television (TV) spectrum with the help of cooperative TV sets. Developed a technical solution and analyzed how much TV-band spectrum could be opportunistically accessed in 274 U.S. cities, based on databases of TV ratings, census, and TV stations [C17, C18, J8]. Developed a dynamic pricing scheme for the network operators to profit and manage congestion under frequent interruptions from TV viewers [J9], and a model of aggregated interference in shared spectrum for interference management [C16].
- 2011–2013 **R&D Testing of 4G TD-LTE Remote Radio Head**, *Ericsson*, references: [P1].
Conducted extensive testing on the radio frontend of Time-Division Long-Term Evolution (TD-LTE) cellular base stations to ensure compliance with internal and regulatory standards in R&D iterations. Diagnosed radio performance issues under various temperature, moisture, and vibration challenges. Designed a patented clock circuit to withstand vibration challenge [P1]. Led a team of 5 engineers to develop in-house test automation software, coaching the entire department in transitioning from manual to Python programming-based testing, resulting in a 10-fold increase in productivity.
- 2009–2011 **Real-time DSP software for 4G WiMAX and LTE Physical Layer**, *ArrayComm*.
Developed real-time digital signal processing (DSP) software for WiMAX and LTE cellular base-stations, including proprietary algorithms for multi-antenna signal processing (beamforming, MIMO, MU-MIMO, channel estimation), and physical layer protocols (MIMO detection, source & channel coding). Optimized software performance with mixed C and Assembly programming. Designed real-time software system on multi/many-core digital signal processors.
- 2006–2009 **Channelized Software-Defined Radio (SDR) Receiver**, *University of Electronic Science and Technology of China*, PI: Zishu He, references: [P2, T2].
Developed a channelized multi-antenna SDR receiver for electronic reconnaissance, in which the input analog intermediate-frequency signal is sampled and down-converted into digital base-band signal, and then transformed into 256 ~ 32768 sub-channels of narrow-band signals via a patented high-speed filter bank [P2]. Contributions include design of mixed-signal printed circuit board (PCB) and DSP system on field-programmable gate array (FPGA).

Academic Service

Conference Activities

- 2025 Organizer and Chair of special session “Machine Learning for Communications and Networking” at Asilomar
- 2023 Chair of session “Machine Learning Applications to Communications” at ICASSP
- 2022 Co-chair of session “Machine Learning for Communications” at ICASSP
- 2020 Member of technical program committee at IEEE VTC’20-Fall

Reviewer Duties

Journals	Reviewer for 22 academic journals	
	IEEE Comm. Surveys & Tutorials	IEEE J. Selected Areas in Comm. (JSAC)
	IEEE Trans. Wireless Comm. (TWC)	IEEE Comm. Letters
	IEEE Trans. Networking	IEEE Trans. ML Comm. & Netw.(TMLCN)
	Journal of Communications and Networks	IEEE Open Journal of Signal Processing
	IEEE Trans. Signal Processing (TSP)	IEEE Signal Processing Letters
	IEEE Trans. Mobile Computing	IEEE Trans. Multimedia
	IEEE Trans. Vehicular Technology (TVT)	IEEE Trans. Cognitive Comm. & Netw.

	IEEE Communications Magazine	IEEE Network Magazine
	IEEE Trans. Signal Info. Process. Netw.	EURASIP J. on Wireless Comm. & Netw.
	Elsevier Physical Communication	The Intl. J. Computer & Telecomm. Netw.
	SCIENTIA SINICA Informationis	Wireless Comm. & Mobile Computing
Conferences	Reviewer for IEEE conferences and AI/ML conferences	
	MILCOM, Asilomar, ICASSP, MLSP, ICC, INFOCOM, GLOBECOM, DySPAN, LCN, SECON, VTC, ICDCS	
	NeurIPS 2025	

Membership

2025–Cont. Association for Computing Machinery (ACM) Member
 2019–Cont. IEEE Signal Processing Society Member
 2013–Cont. IEEE Communications Society Member
 2013–Cont. IEEE Member

Community Service

2017–2018 **Secretary**, *Graduate Student Association*, in Dept. of Computer Science and Engineering, University of Nebraska-Lincoln.

Awards

2025 Best Poster Award, ACM MobiHoc 2025
 2023 Future Faculty Fellow, George R. Brown School of Engineering, Rice University
 2022 IEEE Signal Processing Society Travel Grant for ICASSP 2022
 2006–2009 National Scholarship, University of Electronic Science and Technology of China
 2006 Outstanding Graduate, University of Electronic Science and Technology of China
 2005 First-class prize in National Undergraduate Electronic Design Contests in China
 2003–2005 People’s Scholarship, University of Electronic Science and Technology of China

Skills

Languages Chinese (native speaker), English (fluent)
 Programming Python, Matlab, C/C++, SQL, Bash, Javascript, Assembly, Verilog HDL, R, SAS
 Circuits Mixed-signal and radio-frequency instruments and PCB design

Chartered Financial Analyst (CFA)

CFA[®] Program participant, CFA Institute. Completed CFA level III in 2024. (Max Level III)

Publications

Manuscripts In Preparation

[M1] Zhongyuan Zhao, Yujun Ming, Ananthram Swami, Kevin Chan, Fikadu Dagefu, and Santiago Segarra. Generalizing biased backpressure routing and scheduling to wireless multi-hop networks with advanced air-interfaces. *IEEE Transactions on Networking*, Jan. 2026. in preparation.

Under Review

- [R1] Negar Erfaniantaghvayi, Zhongyuan Zhao, Kevin Chan, Ananthram Swami, and Santiago Segarra. Ant backpressure routing for wireless multi-hop networks with mixed traffic patterns. *Ad Hoc Networks*, pages 1–14, May. 2026. under review.
- [R2] Rostyslav Olshevskyi, Zhongyuan Zhao, Kevin Chan, Ananthram Swami, Olena Tkachenko, and Santiago Segarra. Fully distributed online training of graph neural networks in networked systems. *IEEE Journal on Selected Areas in Communications*, pages 1–13, Dec. 2025. under review.

Journal Articles

- [J1] Zhongyuan Zhao, Gunjan Verma, Ananthram Swami, and Santiago Segarra. Distributed link sparsification for scalable scheduling using graph neural networks. *IEEE Transactions on Wireless Communications*, 25:3879 – 3893, Sep. 2025.
- [J2] Zhongyuan Zhao, Bojan Radojičić, Gunjan Verma, Ananthram Swami, and Santiago Segarra. Biased Backpressure routing using link features and graph neural networks. *IEEE Transactions on Machine Learning in Communications and Networking*, pages 1424 – 1439, Sep. 2024.
- [J3] Eli Chien, Mufei Li, Anthony Aportela, Kerr Ding, Shuyi Jia, Supriyo Maji, Zhongyuan Zhao, Victor Fung, Callie Hao, Yunan Luo, Olgica Milenkovic, David Pan, Santiago Segarra, Javier Duarte, and Pan Li. Exploring the opportunities and challenges of graph neural networks in electrical engineering. *Nature Reviews Electrical Engineering*, 1:529–546, 2024.
- [J4] Brandt Bessell, Josh Loecker, Zhongyuan Zhao, Sara Sadat Aghamiri, Sabyasachi Mohanty, Rada Amin, Tomáš Helikar, and Bhanwar Lal Puniya. COMO: A pipeline for multi-omics data integration in metabolic modeling and drug discovery. *Briefings in Bioinformatics*, 24(6), Nov. 2023.
- [J5] Zhongyuan Zhao, Gunjan Verma, Chirag Rao, Ananthram Swami, and Santiago Segarra. Link scheduling using graph neural networks. *IEEE Transactions on Wireless Communications*, 22(6):3997–4012, June 2023.
- [J6] Zhongyuan Zhao, Mehmet Can Vuran, Fujuan Guo, and Stephen D. Scott. Deep-waveform: A learned ofdm receiver based on deep complex-valued convolutional networks. *IEEE Journal on Selected Areas in Communications*, 39(8):2407–2420, Aug. 2021.
- [J7] Zhongyuan Zhao, Mehmet C. Vuran, Baofeng Zhou, Mohammad M.R. Lunar, Zahra Aref, David P. Young, Warren Humphrey, Steve Goddard, Garhan Attebury, and Blake France. A city-wide experimental testbed for the next generation wireless networks. *Ad Hoc Networks*, 111:102305, Feb. 2021.
- [J8] Zhongyuan Zhao, Mehmet C. Vuran, Demet Batur, and Eylem Ekici. Shades of white: Impacts of population dynamics and tv viewership on available tv spectrum. *IEEE Transactions on Vehicular Technology*, 68(3):2427–2442, March 2019.
- [J9] Demet Batur, Jennifer K. Ryan, Zhongyuan Zhao, and Mehmet C. Vuran. Dynamic pricing of wireless internet based on usage and stochastically changing capacity. *Manufacturing and Service Operations Management*, 21(4):1–21, Feb. 2019.

- [J10] Samil Temel, Mehmet C. Vuran, Mohammad M.R. Lunar, Zhongyuan Zhao, Abdul Salam, Ronald K. Faller, and Cody Stolle. Vehicle-to-barrier communication during real-world vehicle crash tests. *Computer Communications*, 127:172–186, Sep. 2018.
- [J11] Haihong Tang and Zhongyuan Zhao. DSP and CPLD-based digital AC soft starter. *Automation Information*, 2007(5):53–55, May 2007.

Conference Publications

- [C1] Yujun Ming, Zhongyuan Zhao, Fikadu Dagefu, Justin Kong, Terrence Moore, Kevin Chan, Ananthram Swami, and Santiago Segarra. Biased backpressure routing for multihop wireless networks with heterogeneous interfaces. In *IEEE Military Communications Conference (MILCOM)*, May 2026. accepted for publications.
- [C2] Negar Erfaniantaghvayi, Zhongyuan Zhao, Kevin Chan, Ananthram Swami, and Santiago Segarra. Poster: Sparsity-enhanced lagrangian relaxation (SeLR) for computation offloading at the edge. In *26th International Symposium on Theory, Algorithmic Foundations, and Protocol Design for Mobile Networks and Mobile Computing (MobiHoc)*, Oct. 2025.
- [C3] Zhongyuan Zhao*, Yujun Ming*, Kevin Chan, Ananthram Swami, and Santiago Segarra. A differentiable digital twin of distributed link scheduling for contention-aware networking. In *Asilomar Conference on Signals, Systems, and Computers*, Pacific Grove, CA, Oct. 2025.
- [C4] Narjes Nourzad, Jared Coleman, Zhongyuan Zhao, Bhaskar Krishnamachari, Gunjan Verma, and Santiago Segarra. Actor-twin framework for task graph scheduling. In *The 17th Workshop on Adaptive and Learning Agents (ALA)*, May 2025.
- [C5] Rostyslav Olshevskiy, Zhongyuan Zhao, Kevin Chan, Gunjan Verma, Ananthram Swami, and Santiago Segarra. Fully distributed online training of graph neural networks in networked systems. In *IEEE International Conference on Machine Learning for Communication and Networking (ICMLCN)*, pages 1–6, Barcelona, Spain, May 2025.
- [C6] Zhongyuan Zhao, Jake Perazzone, Gunjan Verma, Kevin Chan, Ananthram Swami, and Santiago Segarra. Joint task offloading and routing in wireless multi-hop networks using biased backpressure algorithm. In *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 1–5, Hyderabad, India, April 2025.
- [C7] Negar Erfaniantaghvayi*, Zhongyuan Zhao*, Kevin Chan, Gunjan Verma, Ananthram Swami, and Santiago Segarra. Ant backpressure routing for wireless multi-hop networks with mixed traffic pattern. In *IEEE Military Communications Conference (MILCOM)*, pages 1174–1179, Washington DC, USA, Oct. 2024.
- [C8] Zhongyuan Zhao, Jake Perazzone, Gunjan Verma, and Santiago Segarra. Congestion-aware distributed task offloading in wireless multi-hop networks using graph neural networks. In *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 8951–8955, Seoul, Korea, April 2024.
- [C9] Zhongyuan Zhao, Gunjan Verma, Ananthram Swami, and Santiago Segarra. Enhanced backpressure routing using wireless link features. In *IEEE International Workshop on Computational Advances in Multi-Sensor Adaptive Processing (CAMSAP)*, pages 271–275, Los Sueños, Costa Rica, Dec. 2023.

- [C10] Zhongyuan Zhao, Bojan Radojičić, Gunjan Verma, Ananthram Swami, and Santiago Segarra. Delay-aware Backpressure routing using graph neural networks. In *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 1–5, Rhodes Island, Greece, June 2023.
- [C11] Zhongyuan Zhao, Ananthram Swami, and Santiago Segarra. Graph-based deterministic policy gradient for repetitive combinatorial optimization problems. In *The 11th International Conference on Learning Representations (ICLR)*, pages 1–21, Kigali, Rwanda, May 2023.
- [C12] Zhongyuan Zhao, Ananthram Swami, and Santiago Segarra. Distributed link sparsification for scalable scheduling using graph neural networks. In *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 5308–5312, Singapore, Singapore, May 2022.
- [C13] Zhongyuan Zhao, Gunjan Verma, Ananthram Swami, and Santiago Segarra. Delay-oriented distributed scheduling using graph neural networks. In *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 8902–8906, Singapore, Singapore, May 2022.
- [C14] Zhongyuan Zhao, Gunjan Verma, Chirag Rao, Ananthram Swami, and Santiago Segarra. Distributed scheduling using graph neural networks. In *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 4720–4724, Toronto, ON, Canada (virtual), June 2021.
- [C15] Zhongyuan Zhao, Mehmet C. Vuran, Zahra Aref, David P. Young, Warren Humphrey, Steve Goddard, Garhan Attebury, Blake France, Baofeng Zhou, and Mohammad M. R. Lunar. A city-wide experimental testbed for next generation wireless networks. In *Proc. Int. Balkan Conference on Communications and Networking (BalkanCom’19)*, Skopje, North Macedonia, June 2019.
- [C16] Zhongyuan Zhao and Mehmet C. Vuran. Modeling aggregate interference with heterogeneous secondary users and passive primary users for dynamic admission and power control in tv spectrum. In *Proc. Int. Balkan Conference on Communications and Networking (BalkanCom’18)*, Podgorica, Montenegro, June 2018.
- [C17] Davis Rempe, Mitchell Snyder, Andrew Pracht, Andrew Schwarz, Tri Nguyen, Mitchel Vostrez, Zhongyuan Zhao, and Mehmet C. Vuran. A cognitive radio TV prototype for effective TV spectrum sharing. In *IEEE International Symposium on Dynamic Spectrum Access Networks (DySPAN)*, pages 1–2, Baltimore, MD, May 2017.
- [C18] Zhongyuan Zhao, Mehmet C. Vuran, Démet Batur, and Eylem Ekici. Ratings for spectrum: Impacts of tv viewership on tv whitespace. In *IEEE Global Communications Conference (GLOBECOM)*, pages 941–947, Austin, TX, Dec. 2014.
- [C19] Hongping Hu and Zhongyuan Zhao. A real-time high resolution image compression system based on ADV212. In *2nd International Congress on Image and Signal Processing (CISP)*, pages 1–4, Tianjin, China, Oct. 2009.

Tutorials

- [S1] Santiago Segarra, Ananthram Swami, and Zhongyuan Zhao. Graph-based machine learning in wireless communications. *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, April 2025.

- [S2] Santiago Segarra, Ananthram Swami, and Zhongyuan Zhao. Graph-based machine learning in wireless communications. *IEEE Military Communications Conference (MILCOM)*, Oct 2024.
- [S3] Santiago Segarra, Ananthram Swami, and Zhongyuan Zhao. Graph-based machine learning in wireless communications. *IEEE International Conference on Machine Learning for Communication and Networking (IEEE ICMLCN)*, May 2024.

Patents

- [P1] Zhongyuan Zhao, Weixu Wang, and Luping Pan. PLL and adaptive compensation method in PLL, issued: May 11, 2016. US Patent 9,496,881.
- [P2] Zishu He, Zhongyuan Zhao, Jianzhong Zhang, Ting Chen, and Kexin Jia. Method and apparatus for an implementation of polyphase filter structure, issued: Nov. 14, 2012. China Patent CN101958697B.

Theses

- [T1] Zhongyuan Zhao. *Improving Spectrum Efficiency by Exploiting User and Channel Behaviors for Next Generation Wireless Networks*. PhD dissertation, University of Nebraska-Lincoln, 2019.
- [T2] Zhongyuan Zhao. *Design and Implementation of Channelized Digital Receiver based on PCI-Express*. Master's thesis, University of Electronic Science and Technology of China, 2009.
- [T3] Zhongyuan Zhao. *Design of Motor Soft Starter based on DSP and CPLD*. Bachelor's thesis, University of Electronic Science and Technology of China, 2006.